

COMPLETE MATHEMATICAL LECTURE NOTES ON
METRIC SPACES & MATRIX SPACES

Axioms, Semi-Metrics, Open Spheres, Topological Theorems, Limit Points
& Matrix Norms

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1 Module 1: Foundations of Metric Spaces and Semi-Metrics

A metric space is a mathematical abstraction that formalizes and generalizes the intuitive geometric concept of distance between points in any set.

1.1 Definition of a Metric Space

Definition: Metric and Metric Space

Let X be a non-empty set. A mapping $d : X \times X \rightarrow \mathbb{R}$ is called a **metric** (or distance function) on X if and only if it satisfies the following four fundamental axioms for all elements $x, y, z \in X$:

[M1] **Non-negativity:**

$$d(x, y) \geq 0, \quad \forall x, y \in X \quad (1)$$

[M2] **Identity of Indiscernibles (Definiteness):**

$$d(x, y) = 0 \iff x = y \quad (2)$$

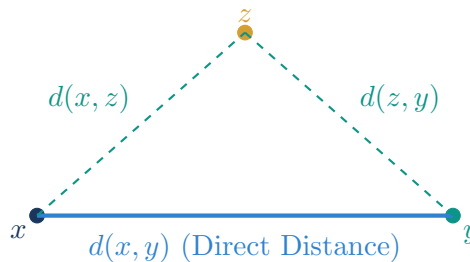
[M3] **Symmetry:**

$$d(x, y) = d(y, x), \quad \forall x, y \in X \quad (3)$$

[M4] **Triangle Inequality:**

$$d(x, y) \leq d(x, z) + d(z, y), \quad \forall x, y, z \in X \quad (4)$$

If d is a metric on X , then the ordered pair (X, d) is called a **Metric Space**, and the non-negative real number $d(x, y)$ is called the **distance between x and y** .



Geometric Axiom [M4]: $d(x, y) \leq d(x, z) + d(z, y)$

1.2 Semi-Metric (Pseudo-Metric)

Definition: Semi-Metric / Pseudo-Metric

A mapping $d : X \times X \rightarrow \mathbb{R}$ is called a **semi-metric** (or **pseudo-metric**) if and only if it satisfies:

$$[\text{M1}] \quad d(x, y) \geq 0, \quad \forall x, y \in X.$$

$$[\text{M2}'] \quad d(x, x) = 0, \quad \forall x \in X \quad (\text{Note: } d(x, y) = 0 \text{ does **not** require } x = y).$$

$$[\text{M3}] \quad d(x, y) = d(y, x), \quad \forall x, y \in X.$$

$$[\text{M4}] \quad d(x, y) \leq d(x, z) + d(z, y), \quad \forall x, y, z \in X.$$

Remark: Every metric is a semi-metric, but a semi-metric is generally **not** a metric because distinct points may have zero distance.

Example: A Semi-Metric that is NOT a Metric

Question: Consider the mapping $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$d(x, y) = |x^2 - y^2|, \quad \forall x, y \in \mathbb{R} \quad (5)$$

Show that d is a semi-metric on $\mathbb{R}$, but is **not** a metric.

Verification of Axioms:

1. **[M1] Non-negativity:** Since absolute value is always non-negative, $d(x, y) = |x^2 - y^2| \geq 0, \forall x, y \in \mathbb{R}$.
2. **[M3] Symmetry:** $d(x, y) = |x^2 - y^2| = |-(y^2 - x^2)| = |y^2 - x^2| = d(y, x)$.
3. **[M4] Triangle Inequality:** For any $x, y, z \in \mathbb{R}$:

$$\begin{aligned} d(x, y) &= |x^2 - y^2| = |(x^2 - z^2) + (z^2 - y^2)| \\ &\leq |x^2 - z^2| + |z^2 - y^2| = d(x, z) + d(z, y) \end{aligned} \quad (6)$$

4. **[M2] Failure of Indiscernibles:**

$$\begin{aligned} d(x, y) = 0 &\iff |x^2 - y^2| = 0 \iff x^2 = y^2 \\ &\iff x = y \quad \text{or} \quad x = -y \end{aligned} \quad (7)$$

For instance, choose $x = 2$ and $y = -2$:

$$d(2, -2) = |2^2 - (-2)^2| = |4 - 4| = 0$$

However, $2 \neq -2$. Therefore, condition $[M2]$ fails.

Conclusion: $d(x, y) = |x^2 - y^2|$ is a **semi-metric** (pseudo-metric), but **not a metric** on $\mathbb{R}$. □

1.3 Standard Metrics on Euclidean Spaces

Standard Metrics on $\mathbb{R}^2$

Let $x = (x_1, y_1)$ and $y = (x_2, y_2)$ be points in $\mathbb{R}^2$.

1. **Euclidean Metric (d_2):**

$$d_2(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (8)$$

2. **Taxicab / Manhattan Metric (d_1):**

$$d_1(x, y) = |x_1 - x_2| + |y_1 - y_2| \quad (9)$$

3. **Maximum / Chebyshev Metric (d_∞):**

$$d_\infty(x, y) = \max(|x_1 - x_2|, |y_1 - y_2|) \quad (10)$$

1.4 Discrete Metric Space

Definition: Discrete Metric Space

Let X be any arbitrary non-empty set. The **discrete metric** on X is defined as:

$$d(x, y) = 0 \quad \text{if } x = y, \quad \text{and} \quad d(x, y) = 1 \quad \text{if } x \neq y \quad (11)$$

The pair (X, d) is called a **Discrete Metric Space**. In this space, every single point is isolated at a distance of 1 from every other point.

1.5 Matrix Spaces as Metric Spaces

Let $M_{m \times n}(\mathbb{R})$ denote the vector space of all $m \times n$ real matrices.

Definition: Frobenius Metric on Matrix Space

For any matrices $A, B \in M_{m \times n}(\mathbb{R})$, the **Frobenius distance** is defined by:

$$d_F(A, B) = \|A - B\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n (a_{ij} - b_{ij})^2} = \sqrt{\text{Tr}((A - B)^T(A - B))} \quad (12)$$

Properties:

- $d_F(A, B) \geq 0$ and $d_F(A, B) = 0 \iff A = B$.
- $d_F(A, B) = d_F(B, A)$.
- $d_F(A, B) \leq d_F(A, C) + d_F(C, B)$ (follows from Cauchy-Schwarz on matrix trace).
- $(M_{m \times n}(\mathbb{R}), d_F)$ is isometrically isomorphic to the Euclidean space $(\mathbb{R}^{mn}, d_2)$ and is therefore a **complete metric space**.

2 Module 2: Open Spheres, Neighborhoods, and Interior Points

2.1 Open Spheres (Open Balls)

Definition: Open Sphere / Open Ball

Let (X, d) be a metric space, $x_0 \in X$, and $r > 0$ be a positive real number. The **open sphere** (or open ball) centered at x_0 of radius r is the subset:

$$S_r(x_0) = B(x_0, r) = \{x \in X \mid d(x, x_0) < r\} \quad (13)$$

The **closed sphere** centered at x_0 of radius r is:

$$\bar{S}_r(x_0) = \bar{B}(x_0, r) = \{x \in X \mid d(x, x_0) \leq r\} \quad (14)$$

2.2 Geometric Shapes of Open Spheres in Different Metrics

Examples: Open Spheres with $x_0 = (0, 0)$ and $r = 1$

1. **Usual Metric on $\mathbb{R}$ with $x_0 = 0, r = 1$:**

$$S_1(0) = \{x \in \mathbb{R} \mid d(x, 0) < 1\} = \{x \in \mathbb{R} \mid |x| < 1\} = (-1, 1) \quad (\text{Open Interval})$$

2. **Euclidean Metric $(\mathbb{R}^2, d_2)$ with $x_0 = (0, 0), r = 1$:**

$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid \sqrt{x^2 + y^2} < 1\} = \{(x, y) \mid \mathbf{x}^2 + \mathbf{y}^2 < 1\} \quad (\text{Circular Disk})$$

3. **Taxicab Metric $(\mathbb{R}^2, d_1)$ with $x_0 = (0, 0), r = 1$:**

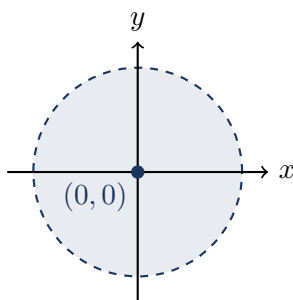
$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid |x| + |y| < 1\} \quad (\text{Rotated Square / Diamond})$$

4. **Maximum Metric $(\mathbb{R}^2, d_\infty)$ with $x_0 = (0, 0), r = 1$:**

$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid \max(|x|, |y|) < 1\} = (-1, 1) \times (-1, 1) \quad (\text{Axis-aligned Square})$$

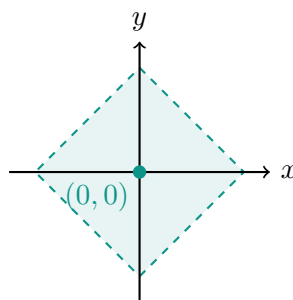
5. **Discrete Metric Space (X, d) :**

$$S_r(x_0) = \{x_0\} \quad \text{if } r \leq 1, \quad \text{and} \quad S_r(x_0) = X \quad \text{if } r > 1$$



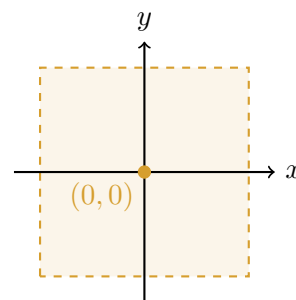
Euclidean Ball (d_2)

$$x^2 + y^2 < 1$$



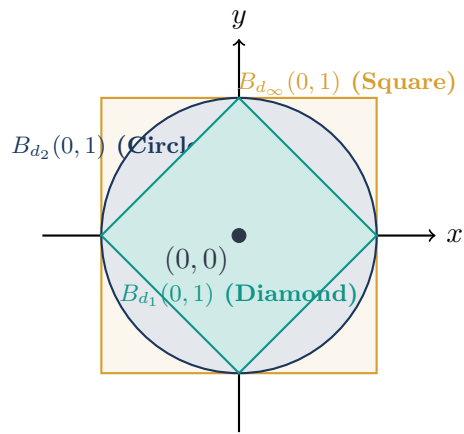
Taxicab Ball (d_1)

$$|x| + |y| < 1$$



Maximum Ball (d_∞)

$$\max(|x|, |y|) < 1$$



Inclusion Hierarchy: $B_{d_1}(0,1) \subset B_{d_2}(0,1) \subset B_{d_\infty}(0,1)$

2.3 Neighborhoods and Interior Points

Definitions: Neighborhood and Open Neighborhood

Let (X, d) be a metric space and $x \in X$.

- A subset $N \subseteq X$ is called a **neighborhood of x** if and only if there exists an open sphere $S_r(x)$ centered at x with radius $r > 0$ such that:

$$S_r(x) \subseteq N \quad (15)$$

- If the neighborhood N is itself an open set, it is called an **open neighborhood** of x .

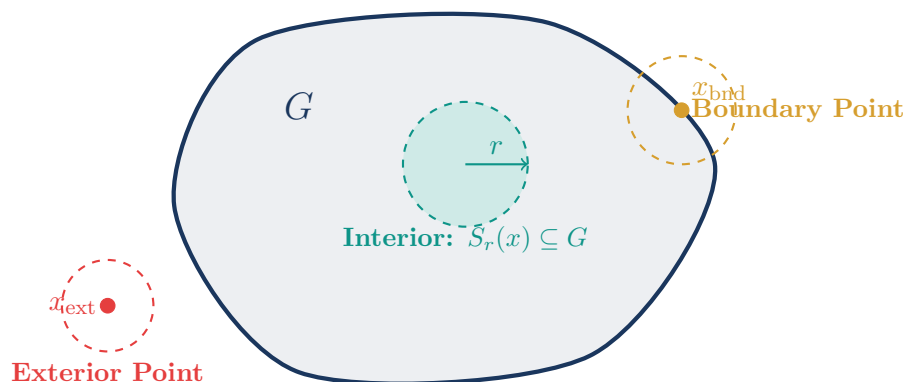
Definitions: Interior Point and Interior of a Set

Let G be a subset of a metric space (X, d) .

- A point $x \in G$ is called an **interior point** of G if and only if there exists an open sphere $S_r(x)$ with $r > 0$ such that:

$$S_r(x) \subseteq G \quad (16)$$

- The set of all interior points of G is called the **interior of G** , denoted by G° or $\text{Int}(G)$.
- Theorem:** A subset G is an **open set** if and only if every point of G is an interior point of G ($G = G^\circ$).



3 Module 3: Topology of Open Sets and Fundamental Theorems

3.1 Open Sets in a Metric Space

Definition: Open Set

A subset G of a metric space (X, d) is called an **open set** if and only if for each point $x \in G$, there exists a real number $r > 0$ such that:

$$S_r(x) \subseteq G \quad (17)$$

3.2 Fundamental Theorems on Open Sets with Complete Proofs

Theorem 1: The Empty Set $\emptyset$ and Full Space X are Open

Theorem: In any metric space (X, d) , the empty set $\emptyset$ and the full space X are open sets.

Proof:

1. **For the empty set $\emptyset$:** A set G is open if every $x \in G$ has an open ball contained in G . Since $\emptyset$ contains no elements, the condition holds vacuously. Thus, $\emptyset$ is open.
2. **For the full space X :** Let $x \in X$ be arbitrary. Choose any radius $r > 0$. By definition, the open sphere $S_r(x) = \{y \in X \mid d(y, x) < r\} \subseteq X$. Hence every point of X is an interior point of X , so X is open. $\square$

Theorem 2: Every Open Sphere is an Open Set

Theorem: In a metric space (X, d) , every open sphere $S_r(x_0)$ is an open set.

Proof:

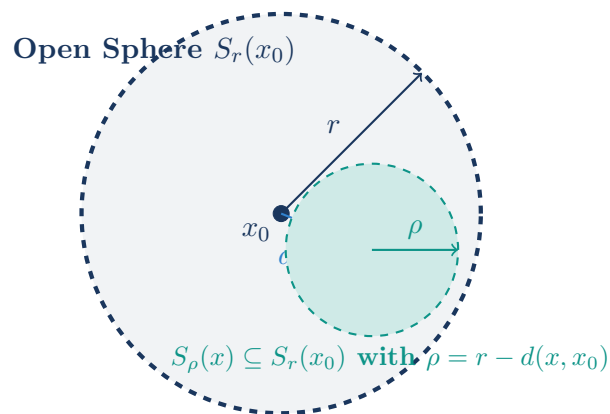
1. Let $S_r(x_0)$ be an open sphere centered at $x_0 \in X$ with radius $r > 0$.
2. Let x be an arbitrary point in $S_r(x_0)$. Then by definition:

$$d(x, x_0) < r \implies r - d(x, x_0) > 0 \quad (18)$$

3. Choose the radius $\rho = r - d(x, x_0) > 0$.
4. **Claim:** $S_\rho(x) \subseteq S_r(x_0)$.
5. To prove the claim, let $y \in S_\rho(x)$. Then $d(y, x) < \rho$.
6. By the triangle inequality [M4] on the metric d :

$$\begin{aligned} d(y, x_0) &\leq d(y, x) + d(x, x_0) \\ &< \rho + d(x, x_0) \\ &= (r - d(x, x_0)) + d(x, x_0) \\ &= r \end{aligned} \quad (19)$$

7. Since $d(y, x_0) < r$, it follows that $y \in S_r(x_0)$.
8. Therefore, $S_\rho(x) \subseteq S_r(x_0)$.
9. Since $x \in S_r(x_0)$ was arbitrary, every point in $S_r(x_0)$ is an interior point.
10. Hence, $S_r(x_0)$ is an **open set**. □



Theorem 3: Arbitrary Union of Open Sets is Open

Theorem: In a metric space (X, d) , the union of an arbitrary collection of open sets is an open set.

Proof:

1. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an arbitrary collection of open sets in (X, d) , where Λ is an arbitrary index set.
2. Let $G = \bigcup_{\lambda \in \Lambda} G_\lambda$.
3. If $G = \emptyset$, then G is open by Theorem 1.
4. If $G \neq \emptyset$, let $x \in G$.
5. By definition of union, $x \in G_{\lambda_0}$ for some specific index $\lambda_0 \in \Lambda$.
6. Since G_{λ_0} is an open set, there exists a radius $r > 0$ such that:

$$S_r(x) \subseteq G_{\lambda_0} \quad (20)$$

7. Since $G_{\lambda_0} \subseteq \bigcup_{\lambda \in \Lambda} G_\lambda = G$, it follows that:

$$S_r(x) \subseteq G \quad (21)$$

8. Since $x \in G$ was chosen arbitrarily, every point of G is an interior point.
9. Hence, $G = \bigcup_{\lambda \in \Lambda} G_\lambda$ is an **open set**. □

Theorem 4: Finite Intersection of Open Sets is Open

Theorem: In a metric space (X, d) , the intersection of a finite number of open sets is an open set.

Proof:

1. Let $G_1, G_2, \dots, G_n$ be a finite collection of open sets in (X, d) .
2. Let $G = \bigcap_{i=1}^n G_i$.
3. If $G = \emptyset$, then G is open by Theorem 1.
4. If $G \neq \emptyset$, let $x \in G$.
5. By definition of intersection, $x \in G_i$ for every $i \in \{1, 2, \dots, n\}$.
6. Since each G_i is open, there exist positive real radii $r_1, r_2, \dots, r_n > 0$ such that:

$$S_{r_i}(x) \subseteq G_i, \quad \forall i \in \{1, 2, \dots, n\} \quad (22)$$

7. Define $r = \min\{r_1, r_2, \dots, r_n\}$. Since it is the minimum of a **finite** collection of strictly positive numbers, $r > 0$.
8. Since $r \leq r_i$ for each i , we have $S_r(x) \subseteq S_{r_i}(x) \subseteq G_i$ for all $i = 1, 2, \dots, n$.
9. Therefore:

$$S_r(x) \subseteq \bigcap_{i=1}^n G_i = G \quad (23)$$

10. Since $x \in G$ was arbitrary, $G = \bigcap_{i=1}^n G_i$ is an **open set**. $\square$

Important Counterexample: Infinite Intersection of Open Sets May NOT be Open

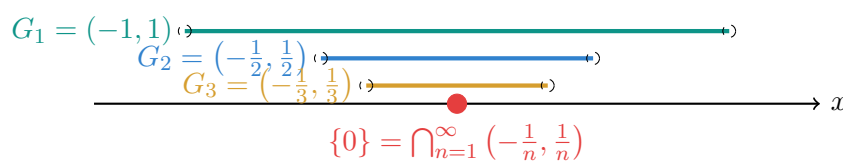
In $(\mathbb{R}, |\cdot|)$, consider the infinite family of open intervals:

$$G_n = \left(-\frac{1}{n}, \frac{1}{n}\right), \quad n \in \mathbb{N} \quad (24)$$

Each G_n is an open set in $\mathbb{R}$. However, their infinite intersection is:

$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\} \quad (25)$$

The singleton set $\{0\}$ is a **closed set** and is **not open** in $(\mathbb{R}, |\cdot|)$, because for any $r > 0$, the open ball $S_r(0) = (-r, r) \not\subseteq \{0\}$.



Infinite sequence of nested open intervals collapsing to closed singleton $\{0\}$

4 Module 4: Limit Points, Derived Sets, and Closed Sets

4.1 Limit Points (Accumulation Points)

Definition: Limit Point of a Set

Let (X, d) be a metric space and $A \subseteq X$. A point $p \in X$ is called a **limit point** (or **accumulation point** / cluster point) of A if and only if every neighborhood N_p of p contains at least one point of A different from p :

$$(N_p \setminus \{p\}) \cap A \neq \emptyset \quad (26)$$

Equivalently, for every $r > 0$:

$$(S_r(p) \setminus \{p\}) \cap A \neq \emptyset \quad (27)$$

4.2 Derived Sets and Examples

Definition: Derived Set A'

The set of all limit points of a subset A is called the **derived set** of A , denoted by A' or $d(A)$.

Solved Examples on Derived Sets

1. **Harmonic Sequence Set in $(\mathbb{R}, |\cdot|)$:** Let $A = \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$. The only accumulation point is 0, because for every $\varepsilon > 0$, $(-\varepsilon, \varepsilon) \cap A$ contains infinitely many terms $\frac{1}{n}$. None of the elements $\frac{1}{n} \in A$ are limit points.

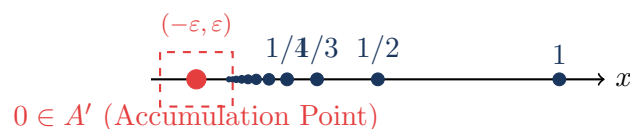
$$A' = \{0\}$$

2. **Open Interval in $(\mathbb{R}, |\cdot|)$:** Let $A = (0, 1)$. Every point in $(0, 1)$ as well as the endpoints 0 and 1 are limit points:

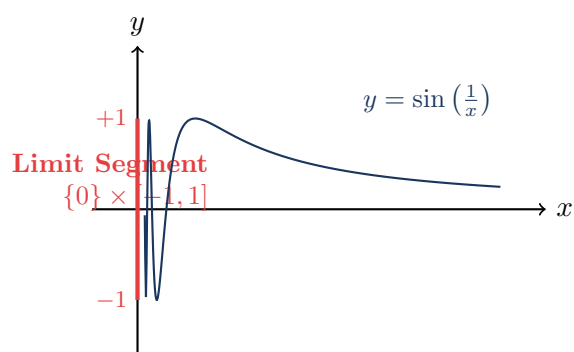
$$A' = [0, 1]$$

3. **Topologist's Sine Curve in $\mathbb{R}^2$:** Consider the subset $A = \{(x, y) \in \mathbb{R}^2 \mid y = \sin(\frac{1}{x}), x > 0\}$. As $x \rightarrow 0^+$, $\sin(1/x)$ oscillates between -1 and $+1$ infinitely often. The derived set A' contains the curve itself plus the entire vertical segment on the y -axis:

$$A' = A \cup (\{0\} \times [-1, 1])$$



Harmonic Set: Points accumulate at 0, so $A' = \{0\}$



Topologist's Sine Curve: Derived Set includes $\{0\} \times [-1, 1]$

4.3 Closed Sets in a Metric Space

Definition: Closed Set

Let (X, d) be a metric space. A subset $F \subseteq X$ is called a **closed set** if and only if its complement $F^c = X \setminus F$ is an open set in (X, d) .

Examples of Closed Sets

1. **Singleton Sets in $(\mathbb{R}, |\cdot|)$:** For any $x \in \mathbb{R}$, the singleton $\{x\}$ is closed because its complement:

$$\{x\}^c = (-\infty, x) \cup (x, \infty)$$

is the union of two open intervals (open sets), hence open. Thus, **every singleton set in a metric space is closed**.

2. **Closed Intervals in $(\mathbb{R}, |\cdot|)$:** $[a, b]$ is closed because $[a, b]^c = (-\infty, a) \cup (b, \infty)$ is open.

4.4 Topological Properties of Closed Sets

Duality Properties of Closed Sets (De Morgan's Laws)

1. The empty set $\emptyset$ and the whole space X are **closed sets** (since $\emptyset^c = X$ and $X^c = \emptyset$ are open).
2. The **intersection of an arbitrary collection** of closed sets is closed:

$$\left(\bigcap_{\lambda \in \Lambda} F_\lambda \right)^c = \bigcup_{\lambda \in \Lambda} F_\lambda^c \quad (\text{Union of open sets is open})$$

3. The **union of a finite number** of closed sets is closed:

$$\left(\bigcup_{i=1}^n F_i \right)^c = \bigcap_{i=1}^n F_i^c \quad (\text{Finite intersection of open sets is open})$$

4.5 Theorem: Limit Point Characterization of Closed Sets

Theorem: Closed Sets Contain All Their Limit Points

Theorem: In a metric space (X, d) , a subset $F \subseteq X$ is closed if and only if F contains all its limit points ($F' \subseteq F$).

Proof:

1. ($\implies$) **Necessary Condition:** Assume F is closed. We must show $F' \subseteq F$.
 - Let $p \in F^c = X \setminus F$.
 - Since F is closed, F^c is an open set.
 - By definition of an open set, there exists an open ball $S_r(p)$ such that $S_r(p) \subseteq F^c = X \setminus F$.
 - This implies $S_r(p) \cap F = \emptyset \implies (S_r(p) \setminus \{p\}) \cap F = \emptyset$.
 - Therefore, p is **not** a limit point of F ($p \notin F'$).
 - By contrapositive, if $p \in F'$, then $p \in F$. Thus $F' \subseteq F$.
2. ($\impliedby$) **Sufficient Condition:** Assume $F' \subseteq F$. We must show F is closed (i.e., F^c is open).
 - Let $x \in F^c = X \setminus F$.
 - Since $F' \subseteq F$, we have $x \notin F'$.
 - Since x is not a limit point of F , there exists an open sphere $S_r(x)$ such that:

$$(S_r(x) \setminus \{x\}) \cap F = \emptyset$$
 - Since $x \notin F$, we have $S_r(x) \cap F = \emptyset$, which implies $S_r(x) \subseteq X \setminus F = F^c$.
 - Since $x \in F^c$ was arbitrary, every point of F^c is an interior point of F^c .
 - Thus F^c is open, which proves that F is **closed**. □

4.6 Theorem: Every Derived Set in a Metric Space is Closed

Theorem: The Derived Set F' is Closed in (X, d)

Theorem: Let (X, d) be a metric space and let $F \subseteq X$. Then the derived set F' is a closed set in (X, d) (that is, $(F')' \subseteq F'$).

Proof:

1. We need to show that $(F')' \subseteq F'$, which by the limit point characterization means F' is closed.
2. If $(F')' = \emptyset$, then $(F')' \subseteq F'$ holds trivially.
3. Now assume $(F')' \neq \emptyset$ and let $x \in (F')'$. That is, x is a limit point of F' .
4. We must prove that $x \in F'$ (i.e., x is a limit point of F).
5. Let $S_r(x)$ be any open sphere centered at x with arbitrary radius $r > 0$.
6. Since $x \in (F')'$, every open sphere centered at x contains at least one point of F' distinct from x . Thus, there exists a point:

$$y_r \in (S_r(x) \setminus \{x\}) \cap F' \quad (28)$$

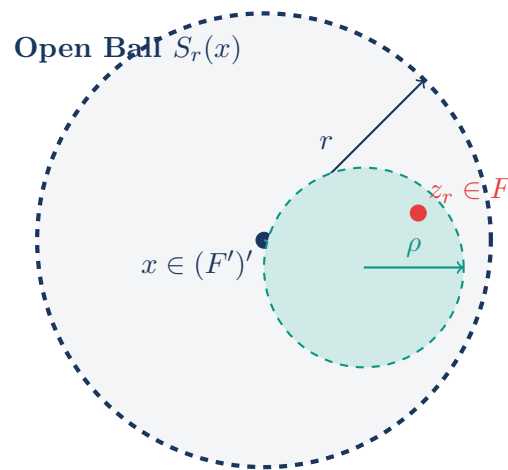
7. Since $y_r \in S_r(x)$, we have $d(x, y_r) < r \implies \rho = r - d(x, y_r) > 0$.
8. Consider the open sphere $S_\rho(y_r)$ centered at y_r with radius ρ .
9. By the Fundamental Theorem on Open Balls, $S_\rho(y_r) \subseteq S_r(x)$.
10. Furthermore, since $y_r \in F'$, y_r is a limit point of F . Therefore, the open sphere $S_\rho(y_r)$ must contain a point $z_r \in F$ distinct from y_r :

$$z_r \in (S_\rho(y_r) \setminus \{y_r\}) \cap F \quad (29)$$

11. Since $z_r \in S_\rho(y_r) \subseteq S_r(x)$, we have $z_r \in S_r(x)$ and $z_r \in F$.
12. By choosing $\rho_1 = \min(\rho, d(x, y_r)) > 0$, we can strictly guarantee that $z_r \neq x$.
13. Therefore, for every $r > 0$, the punctured open sphere $(S_r(x) \setminus \{x\})$ contains a point $z_r \in F$:

$$(S_r(x) \setminus \{x\}) \cap F \neq \emptyset \quad (30)$$

14. This precisely proves that x is a limit point of F , so $x \in F'$.
15. Since $x \in (F')' \implies x \in F'$, we have $(F')' \subseteq F'$.
16. Hence, the derived set F' is a **closed set**. □



Point $z_r \in F$ lies inside $S_\rho(y_r) \subseteq S_r(x)$, proving $x \in F'$

4.7 Closure of a Set and Its Topological Properties

Definition: Closure of a Set ($\bar{A}$)

Let (X, d) be a metric space and $A \subseteq X$. The **closure** of A , denoted by $\bar{A}$ or $\text{Cl}(A)$, is defined as the union of the set A and its derived set A' :

$$\bar{A} = A \cup A' \quad (31)$$

A point $x \in \bar{A}$ is called an **adherent point** (or closure point) of A . That is, every open sphere $S_r(x)$ intersects A ($S_r(x) \cap A \neq \emptyset$).

Theorem: Fundamental Properties of Closure

Let (X, d) be a metric space and $A, B \subseteq X$. The closure operator satisfies:

- (i) $\bar{A}$ is a **closed set** in (X, d) .
- (ii) **Monotonicity:** If $A \subseteq B$, then $\bar{A} \subseteq \bar{B}$.
- (iii) **Characterization:** A is a closed set if and only if $\bar{A} = A$.
- (iv) **Minimality:** $\bar{A}$ is the **smallest closed set** containing A , i.e.,

$$\bar{A} = \bigcap \{F \subseteq X \mid A \subseteq F \text{ and } F \text{ is closed}\}$$

- (v) **Finite Additivity:** $\overline{A \cup B} = \bar{A} \cup \bar{B}$.
- (vi) **Intersection Sub-additivity:** $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$.
- (vii) **Idempotence (Kuratowski Axiom):** $\overline{\bar{A}} = \bar{A}$.

4.7.1 Complete Rigorous Proofs for Closure Properties

Proof of Property (i): $\bar{A}$ is a Closed Set

To Prove: $(\bar{A})^c$ is an open set.

1. By definition, $\bar{A} = A \cup A'$. Using De Morgan's Law:

$$(\bar{A})^c = (A \cup A')^c = A^c \cap (A')^c \quad (32)$$

2. Let $x \in (\bar{A})^c = A^c \cap (A')^c$. Then $x \in A^c$ and $x \in (A')^c$.

3. Since $x \in (A')^c \implies x \notin A'$, x is **not** a limit point of A .

4. Therefore, there exists $r_1 > 0$ such that:

$$(S_{r_1}(x) \setminus \{x\}) \cap A = \emptyset \quad (33)$$

Since $x \in A^c$ ($x \notin A$), this implies $S_{r_1}(x) \cap A = \emptyset \implies S_{r_1}(x) \subseteq A^c$.

5. Moreover, since A' is a closed set (by the previous theorem), its complement $(A')^c$ is an open set.

6. Since $x \in (A')^c$, there exists $r_2 > 0$ such that $S_{r_2}(x) \subseteq (A')^c$.

7. Define $r^* = \min(r_1, r_2) > 0$. Then:

$$S_{r^*}(x) \subseteq S_{r_1}(x) \subseteq A^c \quad \text{and} \quad S_{r^*}(x) \subseteq S_{r_2}(x) \subseteq (A')^c \quad (34)$$

8. Consequently:

$$S_{r^*}(x) \subseteq A^c \cap (A')^c = (\bar{A})^c \quad (35)$$

9. Since $x \in (\bar{A})^c$ was arbitrary, every point of $(\bar{A})^c$ is an interior point.

10. Hence, $(\bar{A})^c$ is open, which proves that $\bar{A}$ is a **closed set**. $\square$

Proof of Properties (ii), (iii), and (iv)**Proof of Property (ii)** $[A \subseteq B \implies \overline{A} \subseteq \overline{B}]$:

1. Let $x \in A'$. For every $r > 0$, $(S_r(x) \setminus \{x\}) \cap A \neq \emptyset$.
2. Since $A \subseteq B$, we have $(S_r(x) \setminus \{x\}) \cap A \subseteq (S_r(x) \setminus \{x\}) \cap B$.
3. Thus $(S_r(x) \setminus \{x\}) \cap B \neq \emptyset$, which means $x \in B'$. Hence $A' \subseteq B'$.
4. Therefore:

$$\overline{A} = A \cup A' \subseteq B \cup B' = \overline{B} \quad \square \quad (36)$$

Proof of Property (iii) $[A \text{ is closed} \iff \overline{A} = A]$:

1. $(\implies)$ Suppose A is closed. Then A contains all its limit points, so $A' \subseteq A$.

$$\overline{A} = A \cup A' = A \quad (37)$$

2. $(\impliedby)$ Conversely, suppose $\overline{A} = A$.
3. Since $\overline{A}$ is always a closed set (by Property (i)), it immediately follows that A is closed. $\square$

Proof of Property (iv) $[\overline{A} \text{ is the smallest closed set containing } A]$:

1. By Property (i), $\overline{A}$ is a closed set, and clearly $A \subseteq A \cup A' = \overline{A}$.
2. Let F be any closed set containing A ($A \subseteq F$).
3. By the monotonicity property (ii), $A \subseteq F \implies \overline{A} \subseteq \overline{F}$.
4. Since F is closed, by Property (iii), $\overline{F} = F$.
5. Therefore, $\overline{A} \subseteq F$.
6. Since this holds for every closed superset F , $\overline{A}$ is indeed the **smallest closed set containing** A . $\square$

Proof of Properties (v), (vi), and (vii)

Proof of Property (v) $[\overline{A \cup B} = \overline{A} \cup \overline{B}]$:

1. Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, by monotonicity (ii):

$$\overline{A} \subseteq \overline{A \cup B} \quad \text{and} \quad \overline{B} \subseteq \overline{A \cup B} \implies \overline{A} \cup \overline{B} \subseteq \overline{A \cup B} \quad (1)$$

2. On the other hand, $A \subseteq \overline{A}$ and $B \subseteq \overline{B} \implies A \cup B \subseteq \overline{A} \cup \overline{B}$.

3. Since $\overline{A}$ and $\overline{B}$ are closed sets, their union $\overline{A} \cup \overline{B}$ is a closed set containing $A \cup B$.

4. Since $\overline{A \cup B}$ is the **smallest** closed set containing $A \cup B$ (Property (iv)), we must have:

$$\overline{A \cup B} \subseteq \overline{A} \cup \overline{B} \quad (2)$$

5. Combining (1) and (2) gives $\overline{A \cup B} = \overline{A} \cup \overline{B}$. □

Proof of Property (vi) $[\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}]$:

1. Since $A \cap B \subseteq A \implies \overline{A \cap B} \subseteq \overline{A}$, and $A \cap B \subseteq B \implies \overline{A \cap B} \subseteq \overline{B}$.

2. Therefore, $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$. □

3. **Counterexample for Strict Inclusion:** In $(\mathbb{R}, |\cdot|)$, let $A = (0, 1)$ and $B = (1, 2)$.

$$A \cap B = \emptyset \implies \overline{A \cap B} = \overline{\emptyset} = \emptyset$$

However, $\overline{A} = [0, 1]$ and $\overline{B} = [1, 2]$, so $\overline{A} \cap \overline{B} = [0, 1] \cap [1, 2] = \{1\} \neq \emptyset$.

Proof of Property (vii) $[\overline{\overline{A}} = \overline{A}]$:

1. Since $\overline{A}$ is a closed set (Property (i)), applying Property (iii) directly to the closed set $\overline{A}$ yields $\overline{\overline{A}} = \overline{A}$. □

4.8 Perfect Sets

Definition: Perfect Set

Let (X, d) be a metric space. A subset $A \subseteq X$ is called a **perfect set** if and only if:

1. A is a **closed set** ($A' \subseteq A$), and
2. Every point of A is a **limit point** of A ($A \subseteq A'$).

Equivalently, a subset A is perfect if and only if it equals its own derived set:

$$A = A' \quad (38)$$

In other words, a perfect set is a closed set with **no isolated points**.

Examples of Perfect and Non-Perfect Sets

1. **Closed Intervals in $\mathbb{R}$:** Any closed interval $A = [a, b]$ with $a < b$ in $(\mathbb{R}, |\cdot|)$ is perfect because $A' = [a, b] = A$.
2. **The Real Line $\mathbb{R}$ and Euclidean Space $\mathbb{R}^n$:** $\mathbb{R}$ is closed and every real number is a limit point of $\mathbb{R}$, so $\mathbb{R}' = \mathbb{R}$, making it perfect.
3. **The Cantor Ternary Set $\mathcal{C}$:** The standard Cantor middle-thirds set is closed, totally disconnected, contains no isolated points, and $\mathcal{C}' = \mathcal{C}$. Thus, the Cantor set is a **perfect set** with Lebesgue measure 0 and cardinality of the continuum ($\mathfrak{c}$).
4. **Non-Perfect Set Example (Singletons and Discrete Sets):** The singleton $\{x\}$ in $\mathbb{R}$ is closed, but its derived set is empty ($\{x\}' = \emptyset \neq \{x\}$). Hence singletons are **not** perfect sets.

Perfect Set: $A = [a, b] \implies A' = [a, b] = A$



Not Perfect: $A = [a, b] \cup \{c\} \implies A' = [a, b] \neq A$



5 Module 5: Interior Operator, Boundary, and Subspace Topologies

5.1 The Interior Operator and Its Fundamental Properties

Definition: Interior of a Set (A° / $\text{Int}(A)$)

Let (X, d) be a metric space and $A \subseteq X$. The **interior** of A , denoted by A° or $\text{Int}(A)$, is the set of all interior points of A :

$$A^\circ = \text{Int}(A) = \{x \in A \mid \exists r > 0 \text{ such that } S_r(x) \subseteq A\} \quad (39)$$

Theorem: Fundamental Properties of the Interior Operator

Let (X, d) be a metric space and $A, B \subseteq X$.

- (i) $A^\circ \subseteq A$.
- (ii) A° is an **open set** in (X, d) .
- (iii) A is an open set if and only if $A^\circ = A$.
- (iv) If $B \subseteq A$ and B is open, then $B \subseteq A^\circ$.
- (v) A° is the **largest open set** contained in A , i.e.,

$$A^\circ = \bigcup \{G \subseteq X \mid G \subseteq A \text{ and } G \text{ is open}\}$$

- (vi) $(A \cap B)^\circ = A^\circ \cap B^\circ$ and $A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$.
- (vii) $(A^\circ)^\circ = A^\circ$ (Idempotence).

5.1.1 Complete Rigorous Proofs for Interior Properties

Proofs of Interior Properties (i) through (v)

Proof of Property (i) [$A^\circ \subseteq A$]:

- By definition, $A^\circ = \{x \in A \mid \exists r > 0, S_r(x) \subseteq A\}$.
- Thus, if $x \in A^\circ$, then by definition $x \in A$. Therefore, $A^\circ \subseteq A$. $\square$

Proof of Property (ii) [A° is an Open Set]:

1. Let $x \in A^\circ$. By definition of an interior point, there exists $r > 0$ such that $S_r(x) \subseteq A$.
2. We must show that there exists an open ball around x entirely contained in A° .
3. In fact, we claim that $S_r(x) \subseteq A^\circ$.
4. Let $y \in S_r(x)$ be any arbitrary point.
5. Since every open sphere in a metric space is an open set (Theorem 2), there exists $\rho = r - d(x, y) > 0$ such that:

$$S_\rho(y) \subseteq S_r(x) \quad (40)$$

6. Since $S_r(x) \subseteq A$, it follows that $S_\rho(y) \subseteq A$.
7. This means that y is an interior point of A , so $y \in A^\circ$.
8. Since $y \in S_r(x)$ was arbitrary, $S_r(x) \subseteq A^\circ$.
9. Hence, x is an interior point of A° . Since $x \in A^\circ$ was arbitrary, A° is an **open set**. $\square$

Proof of Property (iii) [A is open $\iff A^\circ = A$]:

1. ($\implies$) Suppose A is open. By Property (i), $A^\circ \subseteq A$.
2. Now let $x \in A$. Since A is open, by definition there exists $r > 0$ such that $S_r(x) \subseteq A \implies x \in A^\circ$.
3. Thus $A \subseteq A^\circ$. Combining both inclusions gives $A^\circ = A$.
4. ($\impliedby$) Conversely, suppose $A^\circ = A$.
5. Since A° is open by Property (ii), it immediately follows that A is an open set. $\square$

Proof of Properties (iv) and (v) [A° is the largest open set contained in A]:

1. Let $B \subseteq A$ with B being open.
2. Let $x \in B$. Since B is open, $\exists r > 0$ such that $S_r(x) \subseteq B \subseteq A$.
3. This implies $S_r(x) \subseteq A$, so $x \in A^\circ$. Hence $B \subseteq A^\circ$.
4. Since A° is itself an open set contained in A (Properties (i) and (ii)), and contains any other open set $B \subseteq A$, A° is the **largest open set contained in A** . $\square$

5.2 Duality Between Interior, Closure, and Boundary

Complement Duality Relations

For any subset A of a metric space (X, d) :

$$(A^c)^\circ = (\overline{A})^c \quad \text{and} \quad \overline{A^c} = (A^\circ)^c \quad (41)$$

Boundary (Frontier) of a Set ∂A :

$$\partial A = \text{Bd}(A) = \overline{A} \setminus A^\circ = \overline{A} \cap \overline{A^c} \quad (42)$$

5.3 Metric Subspaces and Relative Topologies

Definition: Metric Subspace (Y, d_Y)

Let (X, d) be a metric space, and let Y be a non-empty subset of X ($Y \subseteq X, Y \neq \emptyset$).

- Define the distance function $d_Y : Y \times Y \rightarrow \mathbb{R}$ by restricting d to Y :

$$d_Y(x, y) = d(x, y), \quad \forall x, y \in Y \quad (43)$$

- Since d satisfies axioms [M1] – [M4] on all of X , the restriction d_Y automatically satisfies all metric axioms on Y .
- The pair (Y, d_Y) (or simply (Y, d)) is called a **Metric Subspace** of (X, d) .

Open Spheres in a Metric Subspace

For any $y_0 \in Y$ and $r > 0$, the open sphere in the subspace (Y, d_Y) is:

$$S_r^Y(y_0) = \{y \in Y \mid d_Y(y, y_0) < r\} = \{y \in X \mid d(y, y_0) < r\} \cap Y = S_r^X(y_0) \cap Y \quad (44)$$

That is, **an open ball in Y is simply the intersection of an open ball in X with Y .**

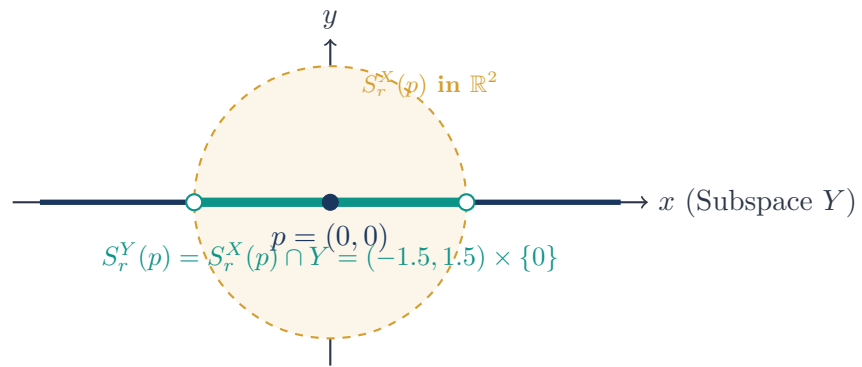
Example: Relative Openness vs. Global Openness

Consider the Euclidean plane $\mathbb{R}^2$ with standard metric d_2 . Let Y be the x -axis:

$$Y = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}$$

Let $E = \{(x, 0) \in \mathbb{R}^2 \mid -1 < x < 1\} = (-1, 1) \times \{0\}$.

- In the subspace (Y, d_Y) , $E = S_1^Y((0, 0))$ is an open ball of radius 1 centered at $(0, 0)$. Thus, **E is open relative to Y .**
- In the full space $(\mathbb{R}^2, d_2)$, every 2D open disk $S_r^{\mathbb{R}^2}((0, 0))$ contains points with non-zero y -coordinates (points off the x -axis). Thus $S_r^{\mathbb{R}^2}((0, 0)) \not\subseteq E$ for any $r > 0$.
- Therefore, **E is NOT open in $\mathbb{R}^2$.**



Relative Open Ball: Open in subspace Y , but 1D line has no interior in $\mathbb{R}^2$

5.4 Fundamental Theorems on Subspace Topology

Theorem: Characterization of Relative Open Sets

Theorem: Let (Y, d) be a metric subspace of (X, d) . A subset $E \subseteq Y$ is **open relative to Y** if and only if there exists an open set $G \subseteq X$ such that:

$$E = Y \cap G \quad (45)$$

Proof:

1. ($\implies$) **Direct Part:** Assume E is open relative to Y .

- For each point $p \in E$, since E is open in (Y, d) , there exists a radius $r_p > 0$ such that the subspace open ball centered at p satisfies:

$$S_{r_p}^Y(p) \subseteq E \quad (46)$$

- Recall that $S_{r_p}^Y(p) = \{q \in Y \mid d(p, q) < r_p\} = S_{r_p}^X(p) \cap Y$, where $S_{r_p}^X(p) = \{q \in X \mid d(p, q) < r_p\}$.
- Let $V_p = S_{r_p}^X(p)$ be the open ball in X . By Theorem 2, each V_p is an open set in (X, d) .
- Define $G = \bigcup_{p \in E} V_p$.
- Since each V_p is open in X and the arbitrary union of open sets is open (Theorem 3), G is an **open set in X** .
- We now verify that $E = Y \cap G$:
 - Since $p \in V_p$ for every $p \in E$, we have $E \subseteq \bigcup_{p \in E} V_p = G$. Since $E \subseteq Y$, it follows that $E \subseteq Y \cap G$.
 - Conversely, let $y \in Y \cap G$. Then $y \in Y$ and $y \in \bigcup_{p \in E} V_p$.
 - Thus $y \in V_{p_0} = S_{r_{p_0}}^X(p_0)$ for some specific $p_0 \in E$.
 - Since $y \in Y$, we have $y \in S_{r_{p_0}}^X(p_0) \cap Y = S_{r_{p_0}}^Y(p_0) \subseteq E$.
 - Hence $y \in E$, which proves $Y \cap G \subseteq E$.
- Combining both inclusions yields $E = Y \cap G$.

2. ($\impliedby$) **Converse Part:** Assume $E = Y \cap G$ for some open set G in (X, d) .

- Let $p \in E = Y \cap G$. Then $p \in Y$ and $p \in G$.
- Since G is open in X , there exists an open ball $S_r^X(p)$ in X with radius $r > 0$ such that $S_r^X(p) \subseteq G$.
- Then the subspace open ball in Y centered at p satisfies:

$$S_r^Y(p) = S_r^X(p) \cap Y \subseteq G \cap Y = E \quad (47)$$

- Thus $S_r^Y(p) \subseteq E$, which proves that p is an interior point of E in the metric space (Y, d_Y) .
- Since $p \in E$ was arbitrary, E is **open relative to Y** . □

Theorem: Characterization of Relative Closed Sets

Theorem: Let (Y, d) be a metric subspace of (X, d) . A subset $K \subseteq Y$ is **closed relative to** Y if and only if there exists a closed set $F \subseteq X$ such that:

$$K = Y \cap F \quad (48)$$

Proof:

1. By definition, K is closed in Y if and only if its relative complement $Y \setminus K$ is an **open set in** Y .
2. By the Relative Open Set Theorem, $Y \setminus K$ is open in Y if and only if there exists an open set $G \subseteq X$ such that:

$$Y \setminus K = Y \cap G \quad (49)$$

3. Taking complements in Y :

$$\begin{aligned} K &= Y \setminus (Y \setminus K) = Y \setminus (Y \cap G) \\ &= Y \cap (X \setminus G) \end{aligned} \quad (50)$$

4. Let $F = X \setminus G = G^c$.
5. Since G is an open set in X , its complement $F = X \setminus G$ is a **closed set in** X .
6. Therefore, $K = Y \cap F$ for the closed set $F \subseteq X$.
7. Conversely, if $K = Y \cap F$ for some closed set $F \subseteq X$, then:

$$Y \setminus K = Y \setminus (Y \cap F) = Y \cap (X \setminus F) = Y \cap G \quad (51)$$

where $G = X \setminus F$ is open in X .

8. Thus $Y \setminus K$ is open in Y , which means K is **closed relative to** Y . □

Important Corollaries on Subspace Open/Closed Sets

1. **When Y is Open in X :** If the subspace Y is an open subset of X , then a subset $E \subseteq Y$ is open in Y if and only if E is **open in** X .
2. **When Y is Closed in X :** If the subspace Y is a closed subset of X , then a subset $K \subseteq Y$ is closed in Y if and only if K is **closed in** X .

6 Module 6: Master Quick Revision Cheat Sheet

1. Metric Axioms, Norms, and Standard Metrics

- **Metric Space Axioms** (X, d) : $\forall x, y, z \in X$:
 1. [M1] **Non-negativity**: $d(x, y) \geq 0$.
 2. [M2] **Identity of Indiscernibles**: $d(x, y) = 0 \iff x = y$.
 3. [M3] **Symmetry**: $d(x, y) = d(y, x)$.
 4. [M4] **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.
- **Standard Metrics on $\mathbb{R}^n$** :
 - Euclidean Metric (L_2): $d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
 - Taxicab / Manhattan Metric (L_1): $d_1(x, y) = \sum_{i=1}^n |x_i - y_i|$
 - Maximum / Chebyshev Metric (L_∞): $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$
 - Discrete Metric: $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.
 - Semi-metric (Pseudo-metric): [M1], [M3], [M4] hold, but $d(x, y) = 0 \not\implies x = y$ (e.g., $d(x, y) = |x^2 - y^2|$).
- **Matrix Frobenius Metric on $M_{m \times n}(\mathbb{R})$** : $d_F(A, B) = \|A - B\|_F = \sqrt{\sum_{i,j} (a_{ij} - b_{ij})^2} = \sqrt{\text{Tr}((A - B)^T(A - B))}$.

2. Open Sets, Spheres, and Neighborhoods

- **Open Sphere (Ball)**: $S_r(x_0) = B(x_0, r) = \{x \in X \mid d(x, x_0) < r\}$.
- **Closed Sphere (Ball)**: $\bar{S}_r(x_0) = \bar{B}(x_0, r) = \{x \in X \mid d(x, x_0) \leq r\}$.
- **Interior Point**: $x \in A$ is an interior point of A if $\exists r > 0$ such that $S_r(x) \subseteq A$.
- **Open Set**: $A \subseteq X$ is open if every point of A is an interior point of A (i.e., $A = A^\circ$).
- **Fundamental Theorem on Open Balls**: Every open sphere $S_r(x_0)$ in (X, d) is an open set (choose $\rho = r - d(x, x_0) > 0$).
- **Topological Properties of Open Sets**:
 - $\emptyset$ and X are open sets.
 - The **arbitrary union** of open sets is open: $\bigcup_{\lambda \in \Lambda} U_\lambda$ is open.
 - The **finite intersection** of open sets is open: $\bigcap_{i=1}^n U_i$ is open.
 - Infinite intersections of open sets need not be open (Counterexample: $\bigcap_{n=1}^\infty (-\frac{1}{n}, \frac{1}{n}) = \{0\}$).

3. Limit Points, Closures, Perfect Sets, and Closed Sets

- **Limit Point (Accumulation Point):** $p \in X$ is a limit point of A if $\forall r > 0, (S_r(p) \setminus \{p\}) \cap A \neq \emptyset$.
- **Derived Set A' :** Set of all limit points of A . **Theorem:** Every derived set A' is a closed set $((A')' \subseteq A')$.
- **Closed Set Characterization:** F is closed $\iff F^c$ is open $\iff F' \subseteq F \iff \overline{F} = F$.
- **Closure of a Set $\overline{A} = A \cup A'$:**
 - $\overline{A}$ is closed.
 - $A \subseteq B \implies \overline{A} \subseteq \overline{B}$.
 - $\overline{A}$ is the smallest closed set containing A ($\overline{A} = \bigcap \{F \supseteq A \mid F \text{ closed}\}$).
 - $\overline{A \cup B} = \overline{A} \cup \overline{B}$, and $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.
 - Idempotence: $\overline{\overline{A}} = \overline{A}$.
- **Perfect Set:** A is closed and has no isolated points $\iff A = A'$. (e.g., $[a, b]$, $\mathbb{R}^n$, Cantor Set $\mathcal{C}$).

4. Interior Operator, Boundary, and Subspace Topologies

- **Interior of a Set $A^\circ = \text{Int}(A)$:**
 - $A^\circ \subseteq A$, and A° is an open set.
 - A is open $\iff A^\circ = A$.
 - A° is the largest open set contained in A ($A^\circ = \bigcup \{G \subseteq A \mid G \text{ open}\}$).
 - $(A \cap B)^\circ = A^\circ \cap B^\circ$, and $(A^\circ)^\circ = A^\circ$.
- **Duality & Boundary:**

$$(A^c)^\circ = (\overline{A})^c, \quad \overline{A^c} = (A^\circ)^c, \quad \partial A = \overline{A} \setminus A^\circ = \overline{A} \cap \overline{A^c}$$
- **Metric Subspaces (Y, d_Y) where $d_Y = d|_{Y \times Y}$:**
 - Open Balls in Subspace: $S_r^Y(y) = S_r^X(y) \cap Y$.
 - **Relative Open Set Theorem:** $E \subseteq Y$ is open in $Y \iff \exists \text{ open } G \subseteq X \text{ such that } E = Y \cap G$.
 - **Relative Closed Set Theorem:** $K \subseteq Y$ is closed in $Y \iff \exists \text{ closed } F \subseteq X \text{ such that } K = Y \cap F$.